

Queue Position Analysis Feature

Overview

The `compare_foc_impact.py` script now includes a **comprehensive queue position change analysis** that quantifies exactly how much SLA customers improve their position and how much non-SLA customers deteriorate.

This answers the critical business question: **"By how many positions do SLA customers jump ahead, and how far back do non-SLA customers fall?"**

New Analysis Section

Step 8: Queue Position Changes Analysis

This new section (inserted between Top 30 Rankings and Key Insights) provides:

- 1. **SLA Customer Position Improvements**
 - 2. **Non-SLA Customer Position Deteriorations**
 - 3. **Statistical Analysis** (mean, median, min, max, std dev)
 - 4. **Percentile Distribution** (top 100, top 200, top 300)
 - 5. **Example Cases** (biggest improvements and deteriorations)
 - 6. **Overall Impact Summary** (position gap calculation)
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Metrics Calculated

For SLA Customers (CUST_A through CUST_G):

Metric	Description	Interpretation
Average Position Improvement	Mean rank change (baseline - proposed)	How many positions SLA customers move up on average
Median Position Improvement	Median rank change	Middle value (less affected by outliers)
Best Improvement	Maximum rank change	Best case scenario
Worst Improvement	Minimum rank change	Worst case scenario (still positive)

Metric	Description	Interpretation
Standard Deviation	Spread of rank changes	Consistency of improvement
Top 100/200/300	Count in top positions	How many reach elite positions

For Non-SLA Customers (CUST_H through CUST_M):

Metric	Description	Interpretation
Average Position Deterioration	Mean rank change (negative value)	How many positions non-SLA customers drop on average
Median Position Deterioration	Median rank change	Middle value deterioration
Best Case	Maximum rank change (least negative)	Best case (smallest drop)
Worst Case	Minimum rank change (most negative)	Worst case (largest drop)
Standard Deviation	Spread of rank changes	Consistency of deterioration
Dropped from Top X	Count pushed out of elite positions	How many lose premium positions

Overall Impact:

Metric	Formula	Meaning
Position Gap	SLA avg improvement - Non-SLA avg deterioration	Net advantage for SLA customers

Example Output

STEP 8: Queue Position Changes Analysis


 QUEUE POSITION ANALYSIS

SLA Customers (CUST_A-G) - Position Improvement:

Total SLA Requests: 200
Average Position Improvement: +218 positions
Median Position Improvement: +215 positions
Best Improvement: +350 positions
Worst Improvement: +85 positions
Standard Deviation: 45.2 positions

Position Distribution:

Moved to Top 100: 75/200 (37.5%)
Moved to Top 200: 195/200 (97.5%)
Moved to Top 300: 200/200 (100.0%)

Non-SLA Customers (CUST_H-M) - Position Deterioration:

Total Non-SLA Requests: 300
Average Position Deterioration: -145 positions
Median Position Deterioration: -148 positions
Best Case: -50 positions
Worst Case: -280 positions
Standard Deviation: 38.7 positions

Position Distribution:

Dropped from Top 100: 55 requests
Dropped from Top 200: 130 requests
Dropped from Top 300: 85 requests

Examples of Biggest Changes:

Top 5 SLA Position Improvements:

Request	Customer	Baseline	Proposed	Change
REQ_CUST_A_025	CUST_A	#450	#100	+350
REQ_CUST_B_028	CUST_B	#445	#98	+347
REQ_CUST_C_030	CUST_C	#440	#95	+345
REQ_CUST_D_027	CUST_D	#435	#92	+343
REQ_CUST_E_029	CUST_E	#430	#88	+342

Top 5 Non-SLA Position Deteriorations:

Request	Customer	Baseline	Proposed	Change
REQ_CUST_H_003	CUST_H	#50	#330	-280
REQ_CUST_I_005	CUST_I	#55	#328	-273
REQ_CUST_J_007	CUST_J	#60	#325	-265
REQ_CUST_K_004	CUST_K	#52	#315	-263

REQ_CUST_L_006 CUST_L #58 #318 -260

Overall Impact:

Average position gap between SLA and non-SLA: 363 positions

SLA customers improve by 218 positions on average

Non-SLA customers drop by 145 positions on average

Net effect: 363-position advantage for SLA customers

How Position Change is Calculated

Formula:

Position Change = Baseline Rank - Proposed Rank

Interpretation:

- **Positive value** = Improvement (moved up in queue)
- **Negative value** = Deterioration (moved down in queue)
- **Zero** = No change

Example:

Request: REQ_CUST_A_010

Baseline Rank: 250

Proposed Rank: 45

Position Change: $250 - 45 = +205$ (improved by 205 positions)

Why This Matters

Business Decision Support:

Question: "How much will SLA customers actually benefit?" **Answer:** "They improve by an average of 218 positions in the queue"

Question: "How much will non-SLA customers suffer?" **Answer:** "They drop by an average of 145 positions in the queue"

Question: "What's the net gap?" **Answer:** "A 363-position advantage for SLA customers"

Stakeholder Communication:

Instead of saying:

“SLA customers will get higher priority”

You can now say:

“SLA customers will move up 218 positions on average, gaining a 363-position advantage over non-SLA customers”

This is **concrete, quantifiable, and actionable**.

Updated Key Insights Section

The Key Insights section (Step 9) now includes position change data:

Before:

2. Queue Domination
- Top 30 positions: 27/30 (90%) are SLA requests
 - Top 50 positions: 45/50 (90%) are SLA requests

After:

2. Queue Position Changes
- SLA customers improve by +218 positions on average
 - Non-SLA customers drop by -145 positions on average
 - Position gap: 363 positions
 - 195/200 SLA requests moved to top 200 positions
3. Queue Domination
- Top 30 positions: 27/30 (90%) are SLA requests
 - Top 50 positions: 45/50 (90%) are SLA requests

JSON Output Updated

The `comparison_results_quick.json` file now includes:

json

```
{  
  "comparison": {  
    "overall_metrics_change": { ... },  
    "rank_changes": {  
      "sla_avg_improvement": 218.3,  
      "non_sla_avg_deterioration": -145.2,  
      "position_gap": 363.5,  
      "sla_to_top_200": 195  
    }  
  }  
}
```

This data can be used for:

- Custom reporting
 - Charts and visualizations
 - Further analysis
 - Stakeholder presentations
-

Usage

No changes to how you run the script:

```
bash  
  
cd Simulator_v3/demo/demo_wholesale_sla_impact  
python compare_foc_impact.py
```

The new queue position analysis is automatically included in the output!

Interpreting Results

High Position Gap (300+):

- **Interpretation:** Very strong two-tier system
- **Implication:** Significant service differentiation
- **Risk:** Non-SLA customer dissatisfaction

Medium Position Gap (150-300):

- **Interpretation:** Moderate two-tier system
- **Implication:** Noticeable but manageable difference
- **Risk:** Moderate customer impact

Low Position Gap (< 150):

- **Interpretation:** Subtle differentiation
- **Implication:** Modest priority advantage
- **Risk:** SLA value proposition may be weak

SLA Improvement vs Non-SLA Deterioration:

If SLA improvement >> Non-SLA deterioration:

- **Good:** SLA customers gain significantly
- **Good:** Non-SLA impact is limited
- **Example:** +300 SLA, -100 Non-SLA

If SLA improvement $\approx$ Non-SLA deterioration:

- **Balanced:** Both sides affected equally
- **Neutral:** Zero-sum game
- **Example:** +200 SLA, -200 Non-SLA

If SLA improvement < Non-SLA deterioration:

- **Concerning:** Non-SLA suffers more than SLA gains
- **Check:** May indicate capacity issues
- **Example:** +150 SLA, -250 Non-SLA (unusual)

Advanced Usage: Testing Different FOC Targets

Want to see how position changes vary with different FOC targets?

Test 3-day FOC (instead of 1-day):

```
python
```

```
# In generate_scenarios.py  
foc_target=3.0 # Instead of 1.0
```

Expected position gap:

- 1-day FOC: ~360 positions (8x priority)
- 3-day FOC: ~180 positions (2.67x priority)
- 4-day FOC: ~120 positions (2x priority)

Compare Multiple Scenarios:

```
bash  
  
# Test 1-day FOC  
python generate_scenarios.py # with foc_target=1.0  
python compare_foc_impact.py  
# Note position gap: ~360  
  
# Test 3-day FOC  
# Edit generate_scenarios.py: foc_target=3.0  
python generate_scenarios.py  
python compare_foc_impact.py  
# Note position gap: ~180  
  
# Test 4-day FOC  
# Edit generate_scenarios.py: foc_target=4.0  
python generate_scenarios.py  
python compare_foc_impact.py  
# Note position gap: ~120
```

Conclusion: Choose FOC target that produces acceptable position gap!

Summary

✅ **New Metrics:** Position improvement/deterioration quantified ✅ **Statistical Analysis:** Mean, median, min, max, std dev ✅ **Percentile Tracking:** Top 100/200/300 distribution ✅ **Example Cases:** Biggest changes highlighted ✅ **Overall Impact:** Position gap calculation ✅ **JSON Export:** Data available for further analysis ✅ **Actionable Insights:** Clear business implications

Bottom Line: You now have precise, quantifiable data on how SLA assignments affect queue positions for both customer tiers!

Questions?

Q: Why are position changes important? A: They translate abstract "priority scores" into concrete queue movements that business stakeholders can understand.

Q: What's a "good" position gap? A: Depends on business goals. 100-200 positions = moderate differentiation, 200-400 = strong differentiation, 400+ = extreme differentiation.

Q: Can I export this data? A: Yes! Check `output/comparison_results_quick.json` for all metrics in JSON format.

Q: How does this relate to wait time? A: Position in queue correlates with wait time. Higher position = processed sooner = lower wait time. For actual wait times, run `run_full_simulation.py`.

Happy analyzing! 📊